



THE RESCUE OF *Wahlenbergia larrainii* IN ROBINSON CRUSOE ISLAND, CHILE

Marcia Ricci

Corporación Nacional Forestal (CONAF), 3 Norte 541, Viña del Mar, Chile

&

Lafayette Eaton

Departamento de Ciencias Ecológicas, Facultad de Ciencias, Universidad de Chile, Casilla 653, Santiago, Chile

(Received 2 March 1992; revised version received 24 June 1993; accepted 14 July 1993)

Abstract

A recent survey by CONAF (Chilean National Forestry Service) of the remnants of the native flora of Juan Fernández island discovered two individuals of *Wahlenbergia larrainii* (Campanulaceae), an endemic species last recorded in 1922. Seeds were collected from the one shrub which produced flowers in 1990; both individuals died in this year. Seeds were germinated and grown in the greenhouse and nursery. The reproductive biology of the surviving offspring was studied, and leaves were used for starch gel electrophoresis.

W. larrainii shows no evidence of apomixis, while selfing produces about half as many fertile seeds as does crossing. The results of electrophoresis are incompatible with those expected for self-fertilization, thus raising the possibility of hybridization with *W. fernandeziana*. Efforts to replant the species in its natural habitat will proceed once its integrity has been confirmed.

Keywords: Juan Fernández Archipelago, *Wahlenbergia larrainii*, conservation measures, propagation, electrophoresis.

INTRODUCTION

The extinction of species, which has reached alarming proportions on a global scale (Wilson, 1988), is especially critical where exotic herbivores have been released on islands (Coblentz, 1978; Armstrong, 1982; Usher, 1988). The result of the introduction of goats and rabbits on the Juan Fernández islands has been the inclusion of practically all its endemic plant species on the endangered species list (Sanders *et al.*, 1982; Benoit, 1989).

The Juan Fernández Archipelago is located about 700 km west of continental Chile (33°S, 78°W), and consists of three islands, Robinson Crusoe (4 711 ha, ex-'Masatierra'), Santa Clara (223 ha) (Fig. 1), and

Marinero Alejandro Selkirk (4 464 ha, ex-'Masafuera'). The flora of the archipelago has interested botanists for some time, both because of its diverse origins (South America, Hawaii, New Zealand and others) and because of its high degree of endemism.

The genus *Wahlenbergia* (Campanulaceae) is represented by six species in Chile. Only *W. linarioides* is found in continental Chile, the other five being endemic to the Juan Fernández Archipelago. Of these, *W. fernandeziana*, *W. larrainii* and *W. grahamae* are found only on Robinson Crusoe Island, *W. berteroi* on both Robinson Crusoe and Santa Clara Islands, and *W. masafuerae* only on Marinero Alejandro Selkirk Island.

Wahlenbergia larrainii was first described as *Campanula larrainii* by Colla (1830) from samples collected by Bertero (1830). Subsequently, De Candolle (1835) referred to it as *Wahlenbergia larrainii* but as a synonym of *W. fernandeziana*. Later, Hemsley (1885) and Johow (1896) placed it under *W. fernandeziana*. Skottsberg (1922) made a complete revision of the genus, clarifying its status and restoring it as a true species.

W. larrainii is a woody shrub which grows up to 2 m tall, with partially persistent, light green, lanceolate leaves and completely white flowers, while *W. fernandeziana* is a hemicryptophyte, has wider, dark green leaves and a purple vein in the flowers. Skottsberg (1922) described the species as rare, but noted that it could be found in the shrublands from Puerto Francés to Vaqueria-Juanango (Fig. 1), while *W. fernandeziana* grows on rocky slopes. None of the subsequent botanical expeditions were able to locate individuals of *W. larrainii*, and it was feared extinct, victim of the human intervention which has drastically reduced the natural vegetation (see Discussion for references). However, a recent extensive survey of the island by CONAF (Corporación Nacional Forestal—Chile) discovered two individuals—one adult plant 2 m high in Cerro los Picos, and a 0.5 m juvenile individual in La Torre. Because of the extreme danger of extinction, a conservation programme was undertaken including rather

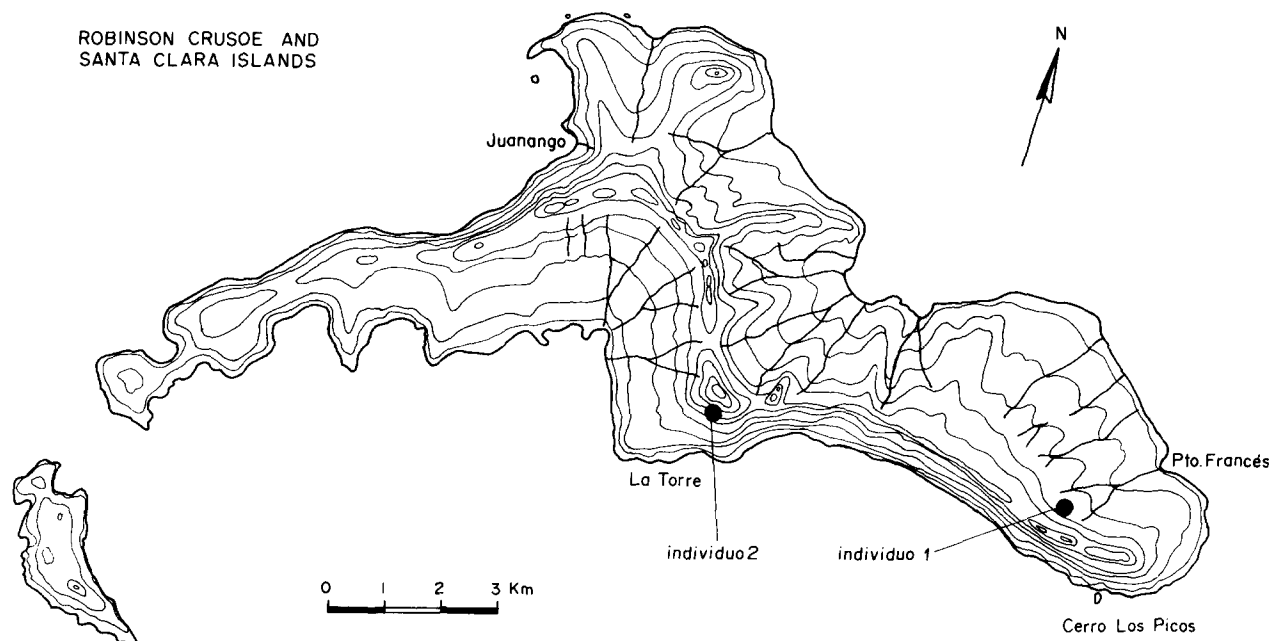


Fig 1. Contour map of Robinson Crusoe and Santa Clara Islands. Contour intervals are 60 m. •, approximate locations of the two individuals of *Wahlenbergia larrainii* discovered by the CONAF survey.

drastic measures to assure the preservation of *W. larrainii*. In this paper, we describe the methods used to rescue and propagate the species, electrophoretic information for the rescued offspring, and information about its biology.

MATERIALS AND METHODS

Search for individuals

The steep slopes of Robinson Crusoe Island (Fig. 1) where the remnants of the native flora persist make the search for plants a difficult and somewhat dangerous enterprise. We made an exhaustive search of the remaining native flora, using permanent transects in those places where it was possible to walk or climb, and lateral searches from these transects using climbing ropes. Although it is still possible that other individuals of the species exist, its conspicuous flowers unfortunately make this unlikely.

Seed collection and propagation

Seeds were collected from the one adult individual found in Cerro Los Picos in March 1989 and were sown in autumn (April) in a greenhouse in Juan Fernández National Park Headquarters on Robinson Crusoe Island, in trays with a mixture of soil and sand. In each of three trays, 100 seeds were placed in a 30 × 30 cm square. These trays were observed daily, recording germination and development of seedlings, for four months. Surviving individuals were transplanted to polyethylene bags and maintained in a nursery.

Phenology and reproductive biology

The phenological changes in the one adult plant discovered were noted monthly, from the time it was found until it died. Those nursery plants that produced at

least eight flowers were used to study aspects of reproductive biology: apomixis, selfing, geitonogamy and cross-fertilization.

Five flowers per plant for each of 10 plants were marked with permanent tags; at appropriate times these were covered with small bags of fine cloth. In one test, each flower was emasculated prior to opening and left covered until the (persistent) petals had dried. In a second test, flowers were simply left covered with bags for their entire life span. In the third test, flowers were emasculated and pollinated the following day with another flower of the same individual, and then covered. The fourth test used the same procedure as the third, but pollination was with flowers of a different plant.

Once mature fruits were produced (when the petals dry), these were harvested and the total number of seeds recorded. Seeds were germinated on moist filter paper at ambient temperature, and the number of fertile seeds noted.

Electrophoresis

Leaves were picked fresh from each offspring, kept on ice during the return to Santiago, and stored at -70° . Samples were ground in 0.2 M phosphate buffer pH 7.5, with addition of bovine albumin (40 mg/100 ml) and 2-mercaptoethanol (2 drops/100 ml buffer), and adsorbed on 2 × 11 mm filter paper wicks. Starch gel electrophoresis was performed in: Tris-citrate pH 8.3 gel/lithium borate pH 8.3 tray (System A of Conkle *et al.*, 1982) for acid phosphatase (ACP), diaphorase (DIAPH), α -esterase (α -EST), glutamate-oxaloacetate transaminase (GOT), glutamate dehydrogenase (GDH), and superoxide dismutase (SOD); 0.005 M histidine pH 8.0 (Brewer, 1970) for alcohol dehydrogenase (ADH), glycerate dehydrogenase (GLY), malic enzyme (ME) and phosphoglucose isomerase (PGI); and morpholine citrate pH 6.1 (System D of Conkle *et al.*, 1982) for

Table 1. Mean number of total seeds and number of fertile seeds per capsule produced by different types of artificial pollination

Type of pollination	Total seeds ($\bar{X}$ + SE)	Fertile seeds ($\bar{X}$ + SE)
Apomixis	0	0
Self-pollination	37.08 + 14.0	18.64 + 7.3
Geitonogamy	47.86 + 11.3	35.88 + 8.6
Cross-pollination	53.98 + 6.0	50.79 + 8.1

aldolase (ALD), hexokinase (HK), isocitrate dehydrogenase (IDH), leucine aminopeptidase (LAP), malic dehydrogenase (MDH), mannose phosphate isomerase (MPI), menadione reductase (MNR), and 6-phosphogluconate dehydrogenase (6-PGD). Staining procedures were from Conkle *et al.* (1982) and Richardson *et al.* (1986).

RESULTS

Search for individuals

After 11 months of searching, one individual of *W. larrainii* was found in February 1989 in Cerro Los Picos at 390 m elevation, on the rocky knife edge of a ridge. In March 1990 a second individual was discovered on an 80° slope on the side of the Cerro El Yunque at 580 m. The second individual showed no evidence of ever having produced flowers. Both these individuals died during 1990, which was the third consecutive dry year in the archipelago.

Seed collection and propagation

The seeds began to germinate after two weeks; within four weeks 95% had germinated. Growth of plantlets was rapid; they had six to eight true leaves by 10–12 weeks after germination. In this stage the plants were transplanted to polyethylene bags. More than half died after transplanting, in spite of the care taken, probably due to the shock of transplantation. Those which survived were maintained for two to three weeks in the greenhouse, and then kept in the nursery.

Phenology and reproductive biology

The only individual of *W. larrainii* in which reproduction could be observed in its native habitat started to flower in the beginning of February, producing the first

seeds at the end of that month. The production of flowers lasted until the middle of April. After this, part of the distal branches dried up and the amount of foliage was considerably reduced. The individuals maintained in the nursery and the experimental garden flowered in their first year, and continued to produce flowers and seeds until the middle of June, after which they lost most of their leaves.

Table 1 shows the results of total seed production and fertile seeds, in the four types of pollination tested. The species does not show evidence of apomixis. Half of the seeds produced by self-pollination were infertile; this percentage is reduced for geitonogamy and is very low for cross-pollination. Both total numbers of seeds produced and numbers of fertile seeds increased from autogamy to cross-fertilization.

Electrophoresis

Five presumptive loci showed variation; their phenotypes are given in Table 2. Where more than one isozyme was present, the most rapidly migrating system was designated as 1, and the most rapidly migrating band was called allele 1 for variable systems. The results are not consistent with those expected for self-pollination for diploid organisms; three alleles were observed for ACP, and three of the other four systems showed genotype distributions significantly different from the 1:2:1 expected for selfing of a heterozygous individual ($\chi^2 + 2$ d.f. with Yates' correction, $p < 0.01$).

The other systems were all invariant (monomorphic). We observed one isozyme in GOT, DIAPH, ME, GDH, a-EST, GLY, HK, MPI, PGI, SOD, and probably IDH; two isozymes in ADH, ALD, 6-PGD, ACP and LAP. Three isozymes were present in MNR. MDH showed five bands of slow migration as in many other plant species—we cannot be sure of the number of isozymes. The number of isozymes observed is less than the usual number for flowering plants (Gottlieb, 1982) for most of the enzymes.

DISCUSSION

W. larrainii was one of the most threatened species of the vascular endemic flora of the Juan Fernández Archipelago when there were only two individuals alive in its native habitat. It may now be extinct in its native range, since both died during 1990, and no others have

Table 2. Electromorphs observed for the five variable systems in 17 offspring of the one adult individual of *W. larrainii* (numbers in the table represent a Mendelian interpretation of the observed bands)

Enzyme	Individual																	Totals of electromorphs			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	11	12	22	23
HK	22	22	22	12	22	22	11	11	22	12	22	22	12	22	22	12	22	2	4	11	
MDH ₁	22	22	22	11	22	22	22	22	22	22	12	22	22	22	12	22	22	1	2	14	
ACP ₁	12	22	22	22	22	22	22	22	23	22	23	12	22	23	23	22	12	0	3	10	4
MNR ₂	11	11	12	22	12	22	12	22	12	22	12	12	22	12	11	22	22	3	7	7	
PGI	22	11	22	22	12	22	11	12	22	22	12	22	22	12	22	22	22	2	4	11	

been found. However, 48 seedlings were found in 1991 at the site of the adult, which we are trying to protect from rabbits.

The drastic decrease in distribution and abundance is notable, comparing the situation found by Skottsberg in 1907 and 1916–17 (Skottsberg, 1922) with the fact that subsequent expeditions were unable to locate any individuals. This change reflects the extreme degree of human intervention to which all the native flora of these islands has been subjected (Philippi, 1856; Johow, 1896; Skottsberg, 1922, 1953a, b; Muñoz-Pizarro, 1969; CONAF, 1976; Mann, 1981; Sanders *et al.*, 1982; Perry, 1984; Hoffman & Marticorena, 1987; Ricci, 1989, 1990).

The seeds collected from the plant in Cerro Los Picos had a high percentage of germination (over 95%), which is important since it is probable that many were the product of autogamy or geitonogamy. Fortunately, both of these mechanisms are relatively successful in this species, producing a mean of 18 and 47 fertile seeds per flower, respectively; otherwise the species might have gone extinct.

The offspring of the single individual do show genetic variability, which is particularly important for the rescue of the species. However, the discovery that not all the offspring are products of selfing is rather startling. There are at least three possibilities to explain these results: (1) there was one (or more) other individual(s) of *W. larrainii* not found in our searches; (2) *W. larrainii* is a functional polyploid; or (3) hybridization occurred with *W. fernandeziana*. Although polyploidy is fairly common in the Campanulaceae (Böcher, 1960), the reduced number of isozymes found does not favour this hypothesis. There is a natural overlap with the flowering period of *W. fernandeziana*, which flowers from the middle of February until the beginning of March, and some of the foliar characters which distinguish the species are not as notable when they are grown together in a common garden, at least in the first 16 months (Ricci, unpublished data).

Crawford *et al.* (1990) studied allozyme variation in three species of *Wahlenbergia*, including *W. fernandeziana*. They found a greater number of isozymes than is usual for plants (Gottlieb, 1982). However, we found fewer isozymes for some systems. We suspect that not all isozymes were sufficiently active in mature leaves to produce staining, perhaps due to inadequate preservation. Crawford *et al.* (1990) suggested that there has been extensive gene duplication or polyploidy in the *Wahlenbergia* of the Juan Fernández Archipelago, but our results do not support this hypothesis for *W. larrainii*.

Several enzymes which were invariant in the present study were variable in *W. fernandeziana* (Crawford *et al.*, 1990); however, we found two alleles for MDH₁, while all MDH isozymes were invariant in the three species studied by Crawford *et al.* (1990). Further information will be necessary to understand the evolutionary relationships among the species of *Wahlenbergia* in the archipelago.

Assuming our Mendelian interpretations are correct,

the strange proportions of electromorphs observed may have resulted partially from selfing, and partially from crossing to an individual homozygous for allele 2 at all five loci (possibly 12 for MNR). It is tempting to use the results to speculate about the proportion of offspring which are selfs, but the high preadult mortality makes this imprudent. We plan to use morphological measures on the offspring, chromosome studies, and electrophoresis with neighbouring individuals of *W. fernandeziana*, in order to attempt to distinguish hybrids from selfs, since the integrity of the species is now in doubt.

We hope to re-establish small populations of these individuals and their offspring *in situ*; we shall use the genetic information to ensure that each group contains as much variation as possible, to provide the greatest opportunity for the species to recover from this maximum bottleneck. Presuming the recolonization is successful, the information on genetic variability in the founding individuals will be invaluable for future studies seeking to explain the mechanisms that maintain or change the frequencies of these genes.

We plan to hybridize the two species, in part to study the magnitude of the differences between them; but we would not release future backcrosses to *W. larrainii* into the natural habitat unless the attempt to re-establish 'pure' *W. larrainii* fails.

ACKNOWLEDGEMENTS

We thank the park guards and personnel of the Juan Fernández National Park, without whose cooperation and sacrifice in the field it would not have been possible to rescue this species from extinction, and Dionisia Sepúlveda for assistance with the electrophoresis. Alberto Veloso, Iván Benoit, César Ormazábal and an anonymous referee made helpful suggestions on the manuscript. Financed in part by CONAF—WWF-US Project No. 3313, 'Conservation and recovery of plants of the flora of Juan Fernández'.

REFERENCES

- Armstrong, P. (1982). Rabbits *Oryctolagus cuniculus* on islands: A case study of successful colonization. *J. Biogeogr.* **9**, 353–62.
- Benoit, I.L. (ed.) (1989). *Libro rojo de la flora terrestre de Chile (primera parte)*. Corporación Nacional Forestal, Santiago, Chile.
- Bertero, C. (1830). Notice sur l'histoire naturelle de l'île Juan Fernández. *Annales des Sciences Natur.*, **XXI**, 344.
- Böcher, T.W. (1960). Experimental and cytological studies on plant species. V. The *Campanula rotundifolia* complex. *Biol. Skr. Dan.*, **11** (4).
- Brewer, G.J. (1970). *An introduction to isozyme techniques*. Academic Press, New York.
- Coblentz, B.E. (1978). The effects of feral goats *Capra hircus* on island ecosystems. *Biol. Conserv.* **13**, 279–86.
- Colla, A. (1835). Plantæ rariores in regionibus Chilensibus a clarissimo M. D. Bertero nuper detectae. *Memorie della Reale Accademia della Scienze di Torino*, **38**, 118.
- CONAF (1976). Plan de manejo del Parque Nacional Archipiélago de Juan Fernández. Documento Técnico, Corporación Nacional Forestal, Santiago, Chile.

- Conkle, M.T., Hodgskiss, P.S., Nunnally, L.B. & Hunter, S.C. (1982). Starch gel electrophoresis of conifer seeds: A laboratory manual. *General Technical Report PSW-54*, Forest Service, US Department of Agriculture.
- Crawford, D.J., Stuessy, T.F., Lammers, T.G. Silva, O.M. & Pacheco, P. (1990). Allozyme variation and evolutionary relationships among three species of *Wahlenbergia* (Campanulaceae) in the Juan Fernández Islands. *Bot. Gaz.*, **151** (1), 119–24.
- De Candolle, A. (1835). *Monographie des Campanulacées*. Veuve Desray, Paris.
- Gottlieb, L. (1982). Conservation and duplication of isozymes in plants. *Science, NY*, **216**, 373–80.
- Hemsley, W.B. (1885). Report on the botany of Juan Fernández and Masafuera. *Report of the Voyage of HMS Challenger. Botany*, **1** (3), 1–96.
- Hoffmann, A. & Marticorena, C. (1987). La vegetación de las islas oceánicas chilenas. In *Las islas oceánicas chilenas; conocimiento científico y necesidades de investigación*, ed. J.C. Castilla. Universidad Católica de Chile, Santiago, Chile, pp. 127–66.
- Johow, F. (1896). *Estudios sobre la flora de las islas de Juan Fernández*. Imprenta Cervantes, Santiago de Chile.
- Mann, G. (1981). Estado actual del plan de manejo del Parque Nacional Archipiélago de Juan Fernández. Documento Interno, Corporación Nacional Forestal, Santiago, Chile.
- Muñoz-Pizarro, C. (1969). El Archipiélago de Juan Fernández y la conservación de sus recursos naturales renovables. *Bol. Acad. Cienc. Chile*, **1**, 83–103.
- Perry, R. (1984). Juan Fernández Islands: A unique botanical heritage. *Environ. Conserv.*, **11**, 72–6.
- Philippi, R.A. (1856). Observaciones sobre la flora de Juan Fernández. *Anal. Univ. Chile*, 157–69.
- Ricci, M. (1989). Programa de conservación y recuperación de las plantas amenazadas de Juan Fernández. Informe final de la primera etapa. Corporación Nacional Forestal, Santiago, Chile.
- Ricci, M. (1990). Programa de conservación y recuperación de las plantas amenazadas de Juan Fernández. Informe final de la segunda etapa. Corporación Nacional Forestal, Santiago, Chile.
- Richardson, B.J., Baverstock, P.R., & Adams M. (1986). *Allozyme electrophoresis*. Academic Press, Sydney.
- Sanders, W., Stuessy, T.F., & Marticorena, C. (1982). Recent changes in the flora of the Juan Fernandez Islands, Chile. *Taxon*, **31**, 284–9.
- Skottsberg, C. (1922). *The phanerogams of the Juan Fernández Islands. Natural history of Juan Fernández and Easter Islands*, Vol. 2, Harqvist, Uppsala, pp. 95–240.
- Skottsberg, C. (1953a). *A supplement to the pteridophytes and phanerogams of Juan Fernández and Easter Islands. Natural history of Juan Fernández and Easter Islands*, Vol. 2. Harqvist, Uppsala, pp. 763–92.
- Skottsberg, C. (1953b). *The vegetation of the Juan Fernández Islands. Natural history of Juan Fernández and Easter Islands*, Vol. 2. Harqvist, Uppsala, pp. 793–960.
- Usher, M.A. (1988). Biological invasion of nature reserves: A search for generalisations. *Biol. Conserv.*, **44**, 119–35.
- Wilson, E.C. (ed.) (1988). *Biodiversity*. National Academy Press, Washington, DC.